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Comparative Study between Quantum and Classical Methods: Few Observations from Portfolio Optimization Problem

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***Abstract*— Classical computers have a limitation in solving NP-hard (nondeterministic polynomial) problems, which cannot be solved in polynomial time. However, the Quantum Computing paradigm is inherently capable of solving such types of problems. One of the major milestones inclined to a quantum -ready era is a relative comparison of execution speed and overall efficiency of current classical versus quantum models. In our approach, we are initiating the foremost step towards addressing the above milestone by performing a comparative study between quantum optimization techniques (available on both gate-model as well as quantum annealers) versus classical by solving a finance domain problem, that is, portfolio optimization using Markowitz mean-variance optimization. Historical market data from Jan'2011 till Dec'2016 of 48 NSE stocks were considered to build the portfolio by optimizing the mean-variance based on Harry Markowitz's Modern Portfolio Theory. We approach this first classically and then on gate-model based quantum computer using Qiskit SDK, followed by D-wave solvers to build efficient portfolio and compare which among these frameworks are mature enough in this NISQ era to solve portfolio optimization problem related to the finance domain. Our results are comparable and from our experimentation performed on D-wave annealers and gate-model simulators, we observed that implementations using quantum methods were faster than the corresponding implementation of classical methods.**

Keywords*— *Quantum Portfolio Optimization, Comparative study Quantum vs Classical, Quadratic unconstrained binary Optimization, Quantum annealing, Gate-model based quantum computing.

I. Introduction

Quantum computing has attracted significant amount of interest in the past few years, particularly in real-world applications, which can offer a significant advantage over the classical one. However, experimental testing of the performance of new algorithms on real-world datasets and comparing them against their classical counterparts through established benchmarks is required to substantiate.

To compare the efficiency of different models supported by different frameworks and to generate the benchmarks, we are utilizing stock market data to build a probabilistic model of a problem in finance domain- portfolio optimization using Markowitz mean-variance optimization. For our study, we have considered daily closing price from historical time-series data corresponding to 48 NSE stocks from 01-01-2011 to 01-11-2021. The benchmark was built against the Nifty50 index Nifty50 index such that, a comparison can be done at the equal Nifty50 index such that, a comparison can be done at the equal ground for both quantum and classical models.

For our quantum approach, we first implement the Variational Quantum Eigen Solver (VQE) and Quantum approximate optimization algorithm (QAOA) on Qiskit. VQE and QAOA are hybrid quantum-classical algorithms implemented in variational form, which use classical optimisers with the quantum machine to solve optimization problems. Secondly, we implement Constrained Quadratic Models (CQM) solvers which is the most advanced solver provided by D-wave's quantum annealing system to find optimal risk versus return portfolio.

Section I, describes the details of the problem identified for our research work. In Section II, details of each computational approaches have been explained. Section III, discusses the main findings of different quantum approaches versus classical model comparison. In Section IV, overall summarization and potential research directions from this work is provided.

II. Problem Description

A portfolio is considered as a suitable assortment of speculations held by an individual or a monetary organization. A portfolio supervisor is obliged to make speculation choices utilizing the cash (or capital) that singular financial backers or monetary establishments have set. An investment perspective on equities raises the following fundamental questions:

- In which of assets should we invest?
- What percentage of budget should we put in each of them?

Hence, finding the optimal investment portfolio is a problem of significant interest in the finance domain. The objective of investors is to invest optimally among a set of assets by respecting some investment constraints. Harry Markowitz's Modern Portfolio Theory aims to determine an optimal portfolio that outcomes the highest expected return (profit) for a defined level of risk (that is, the standard deviation of portfolio returns) or minimizes risk for a given level of expected return.

The goal of portfolio optimisation model is to find the optimal frontier and determine weights for each asset such

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that the total weights sum up to 1. Weight allocation is done to minimize the risk for a given level ρ of the expected return ⟨r(w)⟩. The problem can be written as follows:

$$\min \sum_{i=1}^{N} \sum_{j=1}^{N} \Sigma_{ij}\, w_i w_j \qquad (1)$$
$$\text{s.t. } \langle r(w) \rangle = \sum_{i=1}^{N} \mu_i\, w_i = \rho$$
$$\sum_{i=1}^{N} w_i = 1$$

where, $\sum_{ij}$ specifies the covariances between the assets i and j. μi are the expected returns for the assets.

Portfolio Optimization is a NP-hard problem which is hard to solve in classical computation. Due to the covariance matrix's complex structure, classical algorithms are unable to solve big cases in an acceptable amount of time. The following table (Table I) shows how the computational time increases as the number of assets increases with different combinations. For example, with 8 assets, we want to build a 4 assets portfolio, then the total combinations would be 70 (nCr). To determine the optimal portfolio, for each combination, 10000 iterations are executed. Time taken per loop is 1.6 sec (approx.) , and the total time taken is 9 minutes (approx.). Theoretically, total time increases linearly, and it would take 11000 yrs. (approx.) when we want to build a 10-asset portfolio out of given 50-assets.

TABLE I. COMPUTATIONAL TIME CALCULATION FOR CLASSICAL COMPUTERS

Total No. of Stock	Assets per Portfolio	No. of combination	Iteration per combination	Time Per Loop	Total Time
8	4	70	1000	~ 1.6s	~9 m
8	4	70	10000	~18s	~21m
13	10	286	10000	~35s	~2.5hr
12	4	495	10000	~42s	~6hr
50	10	10272278170	10000	~56s	~11000 yrs.

In prior arts, researchers have either performed a comparative study between classical versus quantum annealers or classical versus gate-model quantum computers. But here, we have taken an extra step to compare classical and quantum annealers and gate model quantum computers using multiple algorithms. Here, our objective is to perform a comparative study between classical and quantum techniques by solving the Portfolio Optimization problem using the Markowitz mean-variance optimization principle.

III. METHODS AND APPROACHES

In this section, we briefly describe the different computational approaches that we have implemented. To implement this, historical time-series market data is used for this study corresponding to 48 NSE stocks for the period encompassing daily close price asset pricing data between 01-01-2011 to 01-11-2021. Data from 01-01-2011 to 31-12-2016 is taken as training data and 01-01-2017 to 01-11-2021 as the validation set.

Symbols / Date	TCS.NS	INFY.NS	HDFC.NS	ICICIBANK.NS	WIPRO.NS	BRITANNIA.NS	AXISBANK.NS	RELIANCE.NS	TATAMOTORS.NS	TITAN.NS
2020-09-01	2246.350098	914.150024	1851.199951	390.950012	271.600006	3771.000000	485.700012	2087.250000	143.800003	1120.050049
2020-09-02	2265.149902	924.000000	1827.199951	392.399994	273.700012	3799.199951	484.399994	2128.199951	150.300003	1121.300049
2020-09-03	2299.500000	935.549988	1808.750000	382.750000	282.850006	3746.850098	474.950012	2112.100098	151.850006	1185.300049
2020-09-04	2288.800049	919.150024	1769.699951	372.549988	276.000000	3705.850098	455.200012	2077.250000	147.800003	1159.250000
2020-09-07	2327.050049	925.049988	1791.099976	373.399994	278.750000	3724.050049	458.700012	2082.649902	149.399994	1167.699951

Fig. 1. Glimpse of the dataset

Finding the Optimal portfolio:

Modern Portfolio Theory is based on mean-variance to find the optimal portfolio for investors. Mainly, there will be the following two parameters considered by investors while choosing the portfolios

1. Maximizes expected return for a given level of risk; and
2. Minimizes risks for a given level of expected returns.

The minimum variance frontier that can be achieved for an expected level of return, can be derived by breaking the process into the following three steps:

1. Mean, variance, and the correlation between each pair of stocks are estimated using the historical data.
2. Stocks iindividual weights are calculated to minimize the portfolio variance against the expected return.
3. Finally, the optimal portfolio is chosen, considering the attributes of maximum return and minimum variance.

Algorithms considered for experimentation rely on Harry Markowitz's Modern Portfolio Theory.

A. The Quantum Learning Pipeline using Qiskit :

1) VQE and QAOA Overview:

VQE and QAOA are the two most popular hybrid quantum-classical algorithms introduced to take advantage of NISQ quantum computers in practical applications.

These algorithms combine a gate-model based quantum computer with a classical optimizer. On the quantum side, the system's initial state is brought into an equal superposition of basis states and evolved by applying the algorithm specific gates to prepare the state which minimizes the cost. A shallow depth characterizes this so-called quantum evolution, also referred to as the ansatz, which enables the algorithm to be run on NISQ devices.

Parametrized Quantum Circuits are used in VQE and QAOA algorithms. Also, uses classical optimizers for solving the problem.

$$\min_{\theta} \langle \psi(\theta) | H | \psi(\theta) \rangle \qquad (2)$$

Through its Hamiltonian H, and more precisely its predicted value, the above equation depicts a system's total energy $\langle \psi(\theta) | H | \psi(\theta) \rangle$. These variational algorithms allow a quantum computer to determine a reasonably optimal solution in polynomial time to a combinatorial optimization problem.

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2) Building the optimal portfolio using VQE and QAOA:

Currently, we have a qubit limitation in using the quantum simulators and real gate-based quantum hardware. For our solution, we used a local state-vector simulator for execution where maximum of 12 qubits were allowed. To overcome that we follow a statistical approach to bucket the 48 assets into 4 buckets containing 12 assets each. This novel bucketing technique makes the solution scalable, enabling to build optimal portfolios out of N number of assets. The bucketing is performed based on the standard deviation of assets, that is, the assets having higher standard deviation are placed in first bucket and subsequently the others are created. This has been completed in such a way that the uncertainty of selecting assets having outliers is minimized and a proper equilibrium is maintained while selecting assets. Initially, using the historical data, the mean, variance of each individual stock in the portfolio, and the correlation of each pair of stocks is calculated. The following plot shows the heat map of one of the 12-asset buckets, that is, a 12X12 matrix:

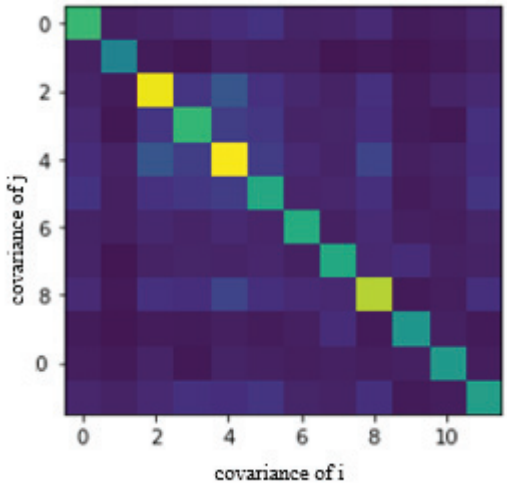

Fig. 2. Heat Map

The left-to-right diagonal values show the relation of a stock with 'itself'. The off-diagonal numbers represent the deviation of each stock's mean expected return from the mean expected return of the other stocks. The following is a straightforward method for examining a covariance matrix.

1. If two stocks either increase or decrease simultaneously, then the covariance value will be positive.
2. If one stock increases while the other stock decreases, then the covariance will be negative.

For applying algorithms such as QAOA and VQE to solve optimization problems our portfolio optimization problem needs to be first converted into a quadratic program. Here, we use Qiskit's finance application modules to convert our portfolio optimization problem into a quadratic program. The quadratic program is formulated using the above defined values.

The portfolio optimization problem in quantum computing can be handled by Quadratic Unconstrained Binary Optimization (QUBO) problems. To find a solution for any QUBO problem is the same as finding the Hamiltonian. A Hamiltonian is an energy function that represents the overall energy of a physical system. For translation of these types of problems, the binary variables accepting the values in the 0,1 binary format are substituted by the spin variables carrying -1, +1 format values. The quadratic program is then converted to Quadratic Unconstrained Binary Optimization (QUBO) problem which is the required format. Next, optimizers are used to optimise the parameters and a solver like VQE and QAOA is called to compute the ground states and return the optimization results. Assets denoted with 1 resembles to be considered for the portfolio and assets denoted with 0 are discarded. After getting the optimised list of assets to be considered for portfolio the weights allocation for the assets are done classically.

Here, in this case, Optimizer is SLSQP (Sequential Least Squares Programming optimizer) [1] and Ansatz is Real Amplitude with 3 repetition layers. Real Amplitude [2] is the circuit consisting of alternating layers of RY rotations and CX entanglements.

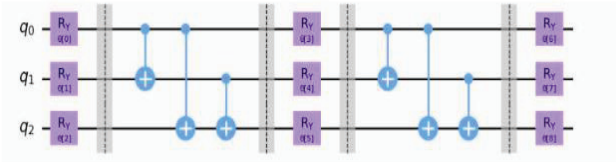

Fig. 3. Sample of Real Amplitude Circuit with 2 repetitions on 3 qubits [2]

Each bucket of 12-assets is passed to the VQE and QAOA algorithms respectively and best 4-assets are being selected from each group. Hence, the resulting 16-asset portfolio is created. In our next section, we will discuss the detailed findings from our experimentation and comparisons of different algorithms.

B. The Quantum Learning Pipeline using D-wave:

1) Quantum Annealing:

Quantum annealing is a metaheuristic that uses quantum tunnelling to determine the global minimum of a given objective function over a set of candidate solutions. Quantum Annealing can be used to solve optimization problems by formulating them as a physical system, where the lowest energy state signifies the optimal solution. In a quantum system, tunnelling can occur, so we may not be stuck in a local minimum.

2) Hybrid CQM Solver Overview:

D-Wave's hybrid solver service (HSS) contains a portfolio of heuristic solvers that leverage quantum and classical methods to solve problems [3]. Constrained Quadratic Models (CQM) is the newest solver provided by D-Wave's hybrid solver service (HSS). Here, we have leveraged this to solve the Portfolio Optimization problem. It is used to solve problems using binary and integer variables that have one or more constraints. In CQM, problem constraints are formulated in the form of inequality and equality. CQM can handle up to 5000 such variables and 100,000 constraints can be created. The constrained quadratic model (CQM) is mathematically defined as follows [4]:

Minimize an Objective:

$$\sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j + c,$$

Subject to constraints:

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$$\sum_i a_i^{(c)} x_i + \sum_{i<j} b_{ij}^{(c)} x_i x_j + c^{(c)} \leq 0,\ c = 1,\ldots,C_{ineq.},$$

$$\sum_i a_i^{(d)} x_i + \sum_{i<j} b_{ij}^{(d)} x_i x_j + c^{(d)} = 0,\ d = 1,\ldots,C_{eq.},$$

where, $\{x_i\}_{i=1,N}$ can be binary or integer variables, a_i, b_{ij}, c are real values and C_{ineq}, $C_{eq.}$ are the number of inequality and equality constraints respectively.

3) Building the optimal portfolio using CQM Hybrid Solver:

Quantum Annealers provided by D-wave can handle up to 5000 qubits, CQM can handle up to 5000 such variables, and 100,000 constraints can be created. Hence, we can provide all the 48 assets together to CQM solver to build the optimal portfolio, unlike the gate-model-based quantum computer provided by Qiskit.

D-wave's Constrained Quadratic Models (CQM) solver is a quantum annealing-based solver which helps achieve global minimum by mapping objective function to the energy state of the system. Problem statement can be divided into two major sections, namely the objective function and the constraints. The objective function is defined as the overall target which needs to be maximized or minimized, whereas constraints define the boundaries in which the solution has to be found. In the case of portfolio optimization problem, the main objective is to maximize returns and minimize risk for a level of risk appetite. The major constraint is that the total portfolio budget should not exceed user-defined budget.

For all 48 NSE stocks, the mean and variance of each of the stocks is calculated. Along with this, a correlation calculation of all the pairs of stocks in also calculated. Budget constraint is set to ensure that the total portfolio budget does not exceed the user-defined budget. The objective function is set and the CQM model is initialised. Then the CQM model is executed using D-wave's LeapHybridCQMSampler which determines the optimal portfolio by reaching the lowest energy state that is obtained from the set of feasible solutions and the weights are calculated internally to the assets in optimal portfolio. In our next section, we will discuss the detailed findings from our experimentation and comparisons of different algorithms.

C. *The Classical Learning Approach:*

Monte Carlo is a stochastic method which uses random sampling of inputs in portfolio optimization, such as, Quantum approaches. In this, same historical data of 48 NSE stocks is considered to calculate the mean and variance of individual stocks as well as the correlation of each stock. The weights were initialised in a uniform way, that is , distributed be from 0 to 1. Then the annual portfolio returns are calculated. by multiplying the average daily returns with the number of trading days, and similarly portfolio variance is also annualized.

Finally, a loop is created which executes N possible (user-defined) simulations and the portfolio having the highest Sharpe ratio is chosen as the optimal one. Sharpe ratio is defined as the measure of performance of an investment such as a security or portfolio compared to a risk-free asset, after adjusting for its risk [5].

IV. Results And Comparisons

For our experiments, we consider as our universe of stocks subsets of historical time-series for 48 NSE stocks using 01-01-2011 to 01-11-2021. Closing price of 48 NSE assets are considered. Data from 01-01-2011 to 31-12-2016 is taken as training data and 01-01-2017 to 01-11-2021 as validation set. A 16-asset portfolio is built out of these 48 stocks using the methods and approaches explained in section II.

The following are the 16-asset portfolios and weights assigned to each asset in portfolio created by the 4 methods explained in section II.

Qiskit VQE		Qiskit QAOA		D-wave CQM		Monte Carlo	
Stock Name	Weights	Stocks	Weights	Stocks	Weights	Stocks	Weights
SHREECEM.NS	12.6%	SHREECEM.NS	9.8%	DIVISLAB.NS	8.01%	SHREECEM.NS	6.1%
BAJAJFINSV.NS	6.2%	BAJFINANCE.NS	12.3%	TCS.NS	19.4%	BAJFINANCE.NS	12.5%
BRITANNIA.NS	12.1%	BRITANNIA.NS	11.5%	HINDUNILVR.NS	3.01%	BRITANNIA.NS	2.9%
DIVISLAB.NS	5.1%	EICHERMOT.NS	14.9%	HDFC.NS	10.0%	EICHERMOT.NS	17.7%
HINDUNILVR.NS	7.1%	HINDUNILVR.NS	0.39%	RELIANCE.NS	9.32%	HINDUNILVR.NS	3.2%
BAJAJ-AUTO.NS	4.1%	KOTAKBANK.NS	0.27%	SUNPHARMA.NS	2.31%	KOTAKBANK.NS	7.2%
HDFC.NS	7.2%	INDUSINDBK.NS	1.89%	CIPLA.NS	2.06%	INDUSINDBK.NS	15.2%
KOTAKBANK.NS	11%	HDFCBANK.NS	11.19%	M&M.NS	5.61%	INFY.NS	8.1%
TECHM.NS	11%	HCLTECH.NS	9.98%	ADANIPORTS.NS	2.50%	TECHM.NS	0.5%
HCLTECH.NS	10%	UPL.NS	0.65%	WIPRO.NS	12.73%	HCLTECH.NS	1.1%
TATASTEEL.NS	2.6%	SUNPHARMA.NS	10.49%	COALINDIA.NS	1.91%	UPL.NS	6.5%
AXISBANK.NS	1.2%	BPCL.NS	8.42%	ITC.NS	10.56%	BPCL.NS	6.6%
ICICIBANK.NS	0.6%	TATAMOTORS.NS	1.19%	ONGC.NS	4.39%	TATAMOTORS.NS	5.4%
WIPRO.NS	1.8%	ITC.NS	3.06%	IOC.NS	5.08%	ITC.NS	1.6%
SBIN.NS	0.4%	IOC.NS	3.41%	POWERGRID.NS	2.01%	IOC.NS	4.02%
ITC.NS	6%	POWERGRID.NS	0.64%	BRITANNIA.NS	1.10%	POWERGRID.NS	0.6%

Fig. 4. Portfolio Weights

Compound Annual Growth is plotted for an interval of 1-year, 3-year and 5-year for the portfolios achieved from 4 different methods and is benchmarked against Index (Nifty).

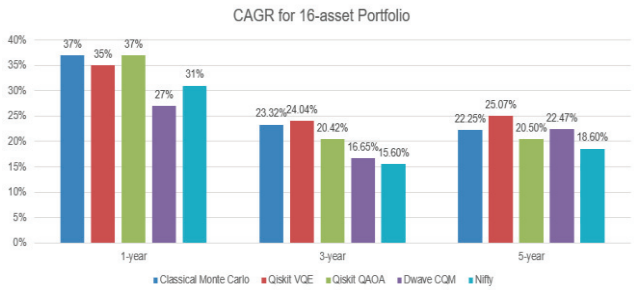

Fig. 5. CAGR Plot

Below table shows the hardware/ simulator details, number of qubits and execution time of the different models:

TABLE II. Model Details

Methods	Simulators / Solvers	No. Of Qubits	Total Execution Time
Qiskit - VQE	State Vector simulator	12 (per bucket)	11 mins
Qiskit - QAOA	State Vector simulator	12 (per bucket)	3.33 mins
D-Wave - CQM	LeapHybridCQM Solver	48	44 Sec
Classical Monte Carlo	NA	NA	16 hrs

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V. SUMMARIZATION AND FUTURE WORK

Understanding whether the available quantum models developed for near-term quantum devices could be used to solve industrial-scale applications is one of the breakthroughs toward a quantum-ready generation. In our work, we compare Qiskit VQE, Qiskit QAOA, Dwave's CQM and classical Monte Carlo as models to solve the portfolio optimization problem in the finance domain.

Although the results achieved from quantum approaches are comparable with the classical one, several research directions could be addressed in future. A natural extension of this work would be to carry out this experimentation using other such models made available by different providers such as, Azure (Stochastic Monte Carlo, Quantum Monte Carlo, and so on) to solve the problems related to the finance domain. Since, we have carried out the experimentation only on D-wave and IBMQ, running on different hardware such as, IONQ, Rigetti, and so on, is another task, including comparative performance and benchmarking.

D-wave provides higher number of qubits (up to 5000 qubits) as compared to the gate-based and annealer based quantum computers. Additionally, in some cases larger problems can be handled in D-wave as compared to gate-based quantum computers, which provides a smaller number of qubits.

From the above CAGR plot, we observe that both classical and quantum approach are providing equally good and comparable results. From our experimentation performed on D-wave annealers and gate-model simulators, we observed that implementations using quantum methods were faster than the corresponding implementation of classical methods

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